29/12/2025 (FN)

Hall Ticket Number:

MR-22

Code: B325CS4

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2025/January 2026

Object Oriented Programming through Java

(Common to CSE, CSB and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Compare and contrast overloading and overriding methods.	1	2	1	1,2
1. b)	What is a constructor? What are its special properties?	1	1	1	1,2
1. c)	Specify the two ways of using super keyword and two advantages of final keyword.	1	2	2	1,2
1. d)	Draw the table which describes between various access specifiers present in Java.	1	2	2	1,2
1. e)	Differentiate between **equals()** method and == while comparing two Strings.	1	3	3	1,2
1. f)	What is the difference between multiprocessing and multithreading?	1	1	3	1,2
1. g)	List the user interface components in event handling.	1	1	4	1,2
1. h)	What is the purpose of event listeners?	1	2	4	1,2
1. i)	How do applets differ from application programs?	1	2	5	1,2
1. j)	What is MVC architecture?	1	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Define a class Motor Vehicle as described below. Data members: (i) modelName (ii) modelNumber (iii) modelPrice Methods: (i) display () method to display the name, price, and model number. Define another class named Car that inherits the class Motor Vehicle and has the following: Data Members: i. discountRate Methods: ii. display () method to display the Car name, Car model number, Car price and the discount rate. Discount () method to compute the discount Create the classes Motor Vehicle and Car with suitable constructors and test it.	5	3	1	1,2
2. b)	What is the need for static keyword? Explain why main method is static. What are the three conditions to be considered when the variable/method is static?	5	2	1	1,2
	OR				
2. c)	Create a class named Employee with the following details: Data members: (a) name (b) address (c) age (d) gender Method: (a) display() to show the employee details Create another class FullTimeEmployee that inherits the Employee class:	5	3	1	1,2

	Data members: (a) salary (b) designation Method: (a) display() to show the salary and designation along with the other employee details Create another class PartTimeEmployee that inherits the Employee class: Data members: (a) workingHours (b) ratePerHour Methods: (a) calculatePay() to calculate the amount payable (b) display() to show the amount payable along with other employee details Create objects of these classes and call their methods. User appropriate constructors.				
2. d)	Write short notes on Garbage Collector.	5	2	1	1,2
3. a)	Define an Abstract class and an Interface. Give the reasons of the situation when to use each of them.	5	2	2	1,2
3. b)	Discuss the types of Inheritance that exist in OOP languages	5	2	2	1,2
	OR				
3. c)	Why Java doesn't support Multiple Inheritance. How can we achieve it with an example?	5	3	2	1,2
3. d)	Define a package in Java. How to create a user defined package in java.	5	2	2	1,2
4. a)	Create a try block that is likely to generate three types of exception and then incorporate necessary catch blocks to catch and handle them appropriately.	5	2	3	1,2
4. b)	Write a program showing the actions from 3 threads. Use Runnable interface to create the threads. Make sure that the main thread always executes last.	5	3	3	1,2
	OR				
4. c)	Consider a Student examination system that prints the mark sheet of students. Input the following from the command line: student's name and marks in six subjects. These marks should be between 0 and 50. If the marks are not in the specified range, raise a Range Exception, else find the total marks and print the percentage of the students.	5	3	3	1,2
4. d)	Describe the complete life cycle of a thread?	5	2	3	1,2
5. a)	Write a java program for creating menu where it allows to choose food in 3 categories Rice, Curry, Starter. When Rice item is selected it will display all types of rice available in the hotel and user can order by selecting type of the rice. Similarly user can select Curry and starter. And display the selected food.	5	4	4	1,2,3
5. b)	Explain different types of layout manger.	5	2	4	1,2
	OR				
5. c)	Write a Java program which has 2 panels. Left panel lists the different programme of the college (CSE, ECE….) in a specific location and the right panel will display the list of faculty details when the user selects one of the programme names.	5	4	4	1,2,3
5. d)	Explain the different mouse and keyboard handling events.	5	2	4	1,2